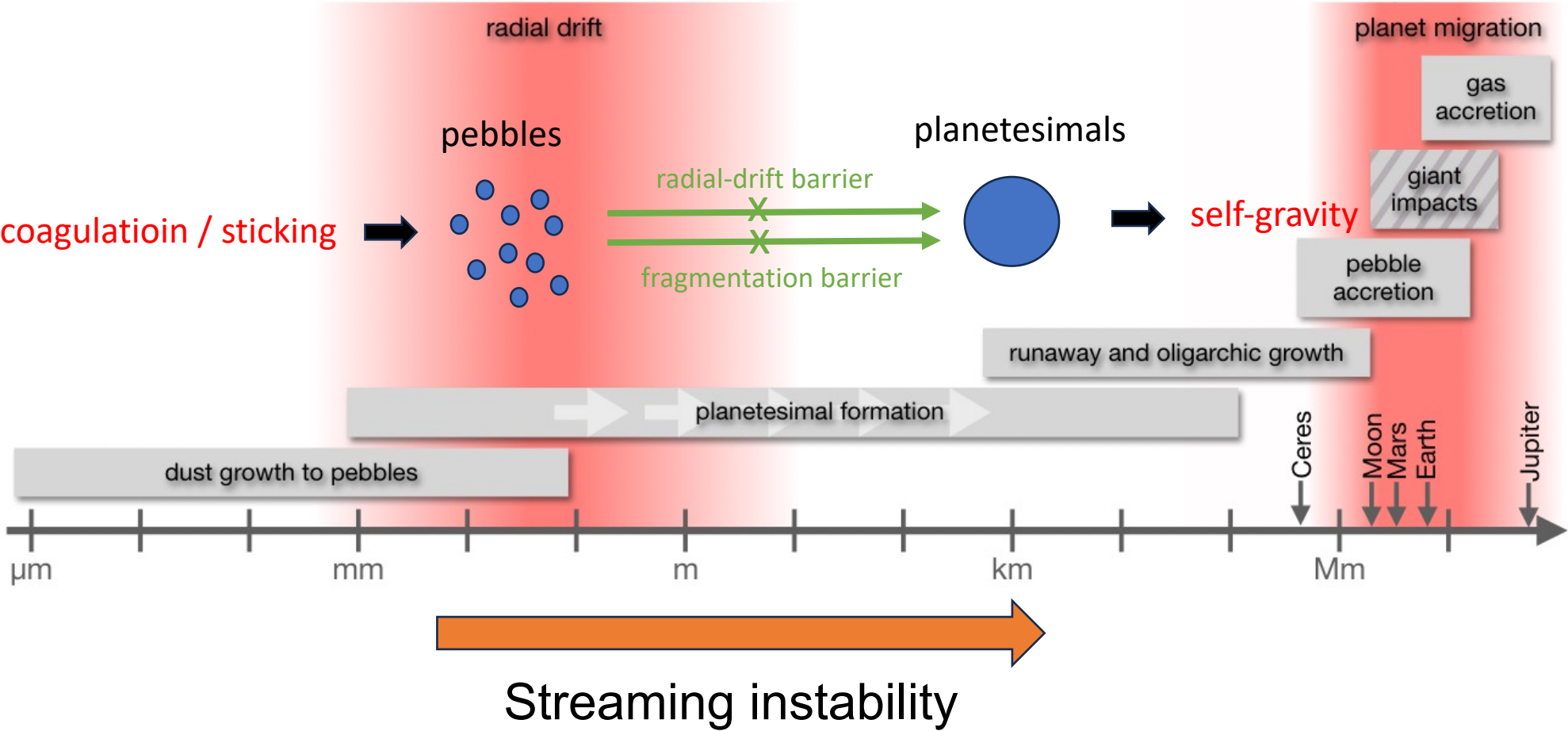


How azimuthal torque can affect dust growth via streaming instabilities

Shiang-Chih Wang (NTHU/ASIAA) & Min-Kai Lin (ASIAA)

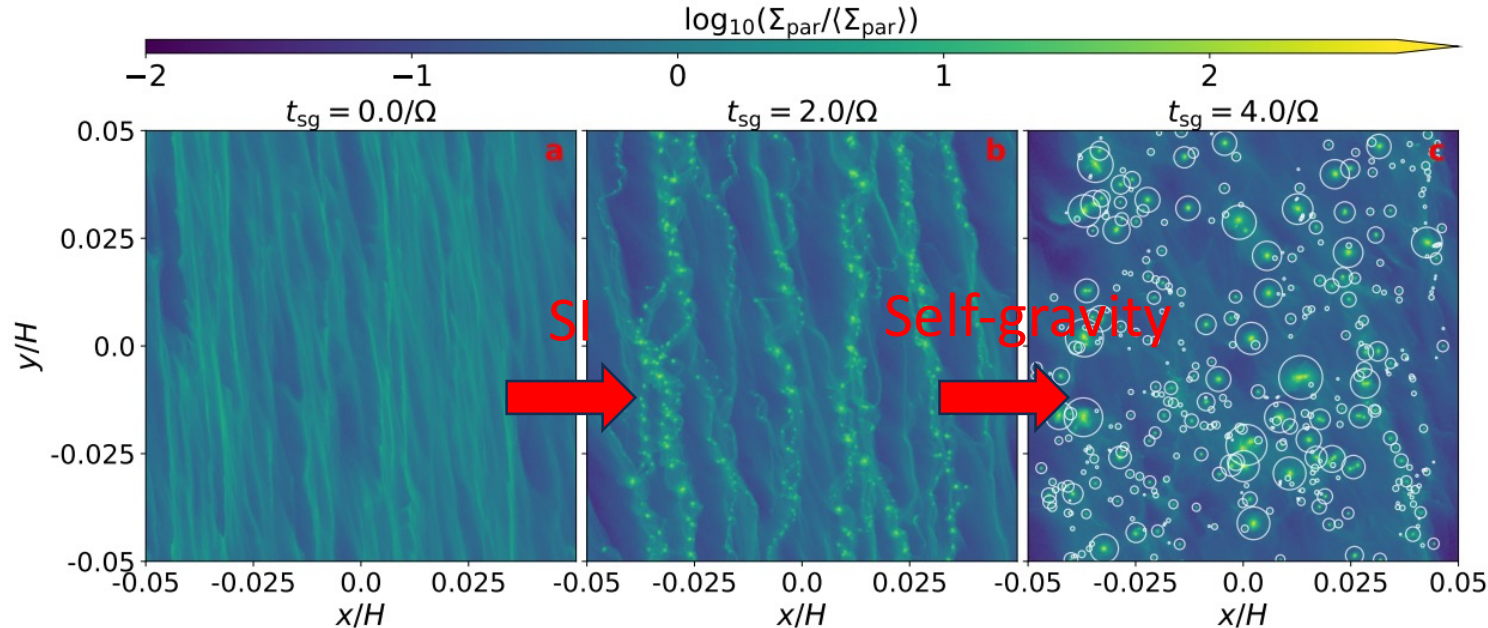


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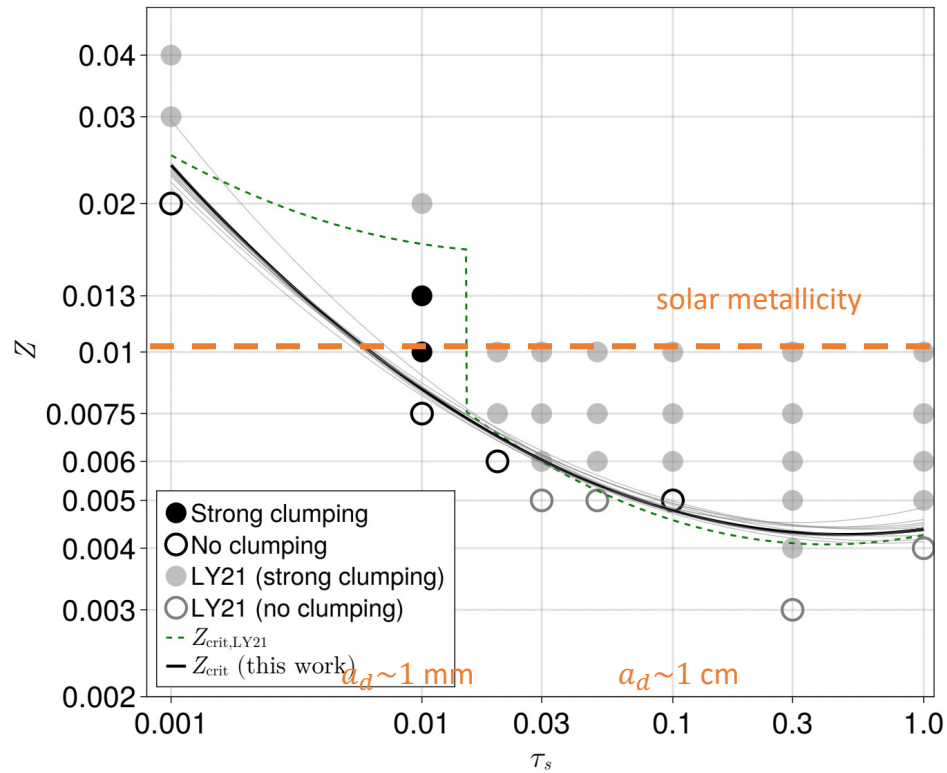
Streaming instability (SI)

- The **radial pressure gradient** $\tilde{\eta}$ provides the necessary background flow for instability.
- Driven by the **relative drift** between the sub-Keplerian gas and Keplerian dust grains.
- Leads to local dust-to-gas ratio enhancement.
- **Gravitational collapse** occurs when local dust density exceeds the critical threshold.



Nevorney et al. 2019

Threshold for planetesimal formation



Lim et al. 2025

Z : Integrated dust-to gas ratio

τ_s : Dynamical stopping time

St : Stokes number

$\tau_s \sim St$ in a Keplerian disk

Z_{crit} was found as the critical boundary for strong clumping.

Minimum-mass solar nebula (MMSN) model (Chiang et al. 2010):

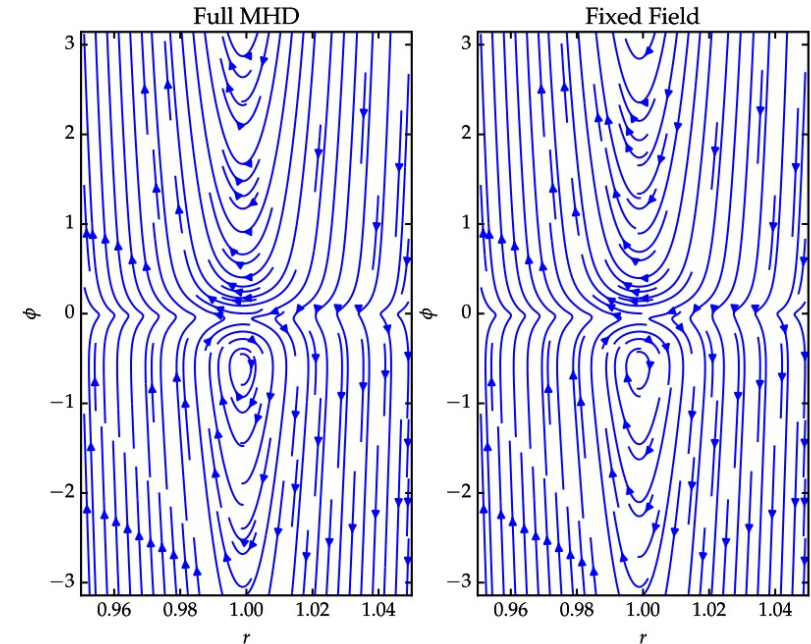
- gas temperature, scale height, and density profile related to radius.
- $St=0.1, 0.01$ corresponds to cm-sized and mm-sized dust grain.

Condition of strong clumping:

- $\rho_d > 100 \rho_g$
- gravitational collapse will happen if dust density exceed clumping condition.

Azimuthal-drift streaming instability (AdSI)

- Hall-induced horizontal magnetic field
- Fixed field is identical to full MHD simulation.
- adding **additional torque** in azimuthal direction.
- Velocity difference in azimuthal direction induces radial accretion flow with strength α_M .
- A new type of SI (AdSI) discovered without requiring a radial pressure gradient. (e.g., Hsu & Lin 2022; Lin & Hsu 2022).
- MMSN model:
 - $\alpha_M = 0.01$ is consistent with standard accretion rate $\dot{M}_* \sim 10^{-8} M_\odot \text{ yr}^{-1}$ for a disk with sun-like central star. (Hartmann et al. 2006)



McNally et al. 2017

Simulation setup

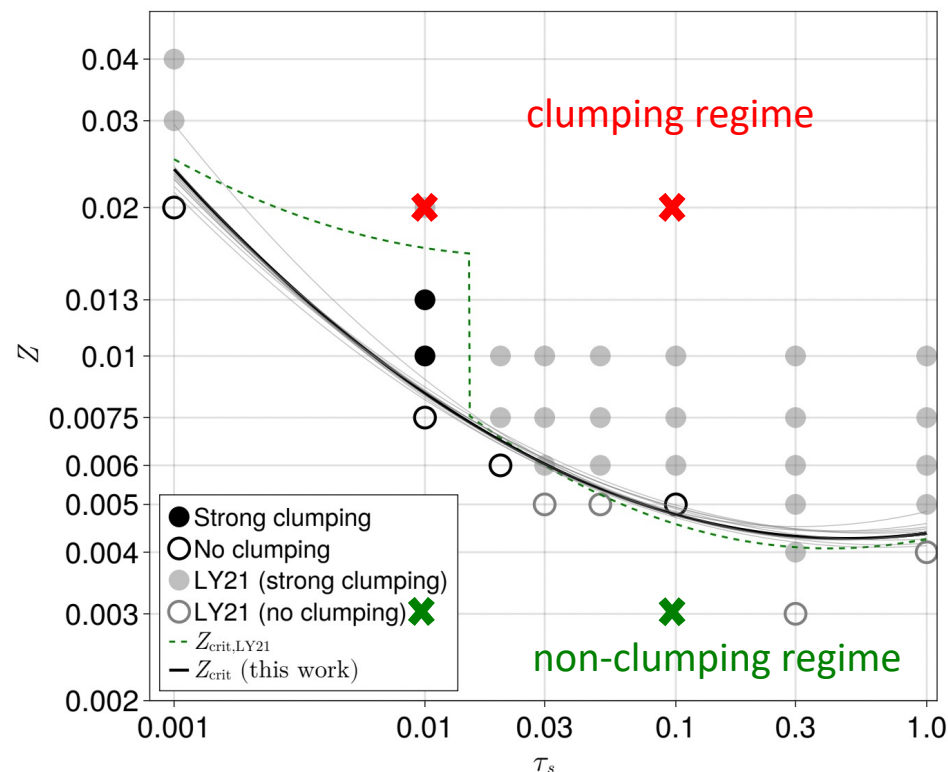
Unstratified disk

- 2D Shearing box simulation using FARGO3D code.
- $St=0.1$
- No self-gravity.
- Vary ϵ and α_M in space $[0.01,100]$ and $[0.001,1]$.

Stratified disk

- 2D Shearing box simulations using Athena++ code, with multifluid dust module (Huang et al. 2022)
- Stratified disk setup, with vertical gravity act on both dust and gas.
- No self-gravity.
- Choosing 2 cases in clumping regime and 2 cases in non-clumping regime.

Simulation setups



Lim et al. 2025

Z : Integrated dust-to gas ratio

τ_s : Dynamical stopping time

St : Stokes number

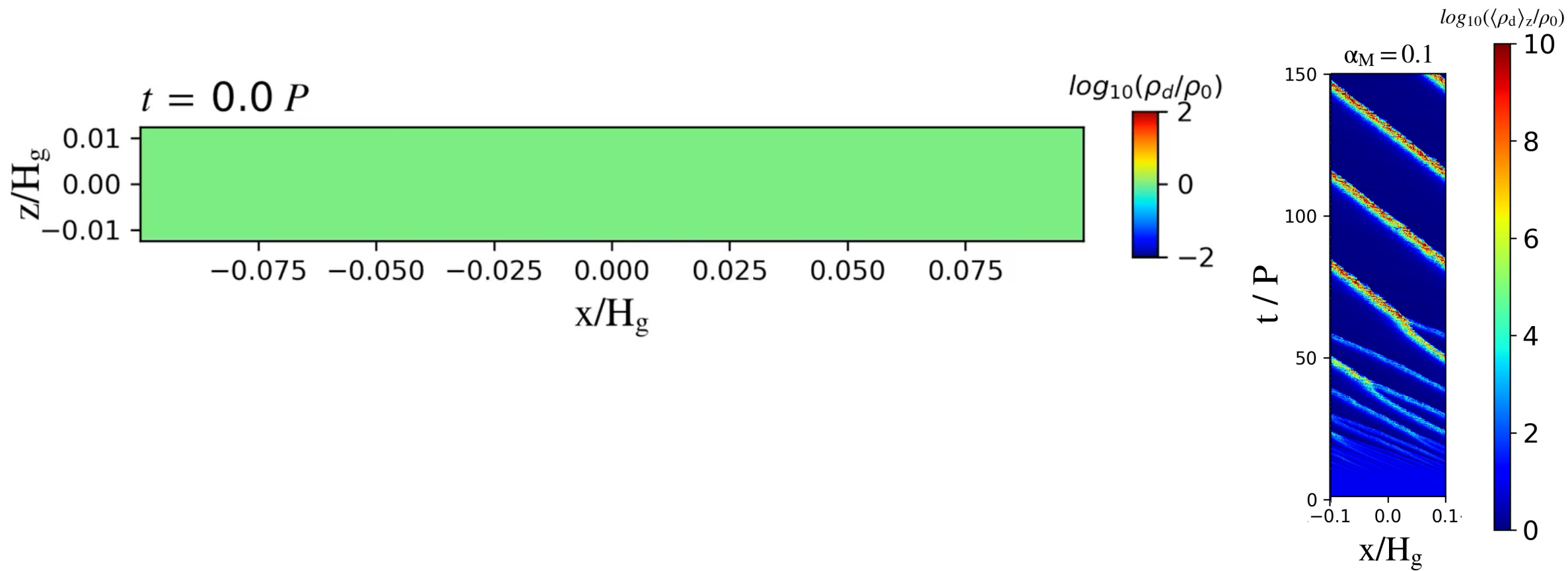
- $(Z, St) = (0.02, 0.1)$ and $(0.02, 0.01)$
- $(Z, St) = (0.003, 0.1)$ and $(0.003, 0.01)$

In clumping regime

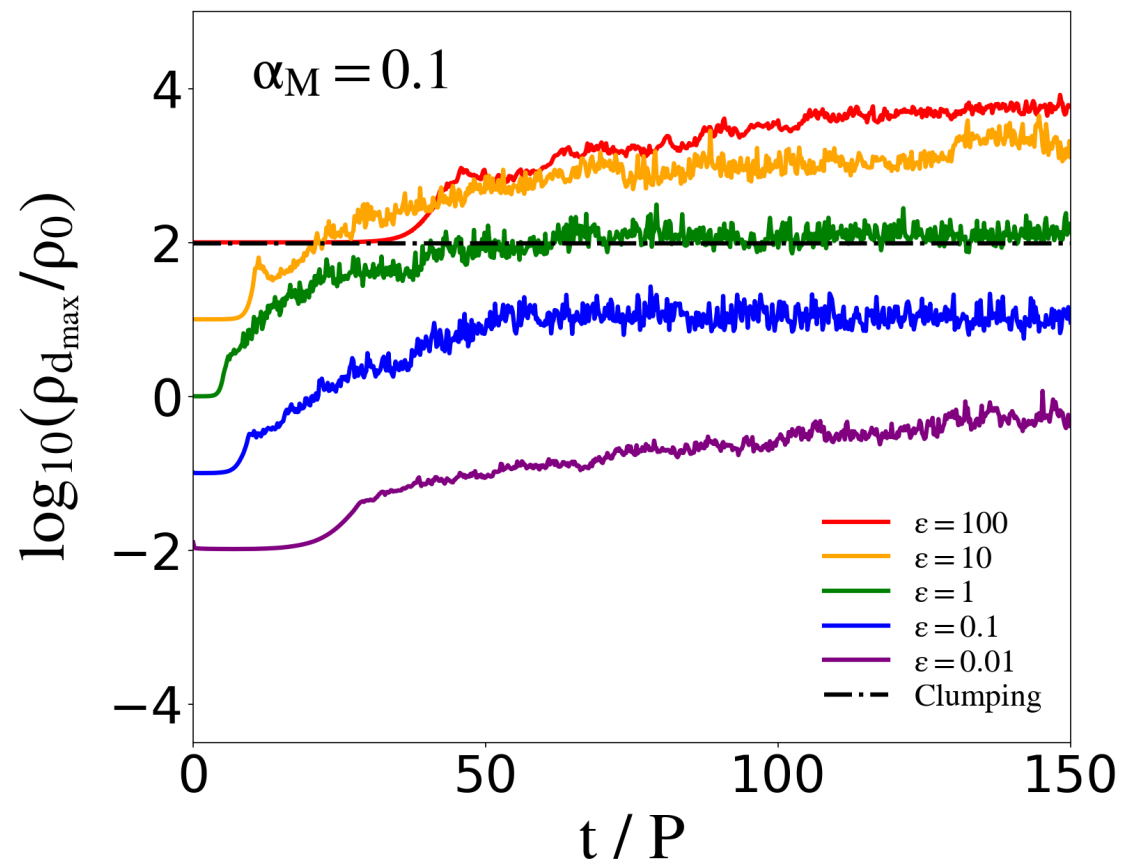
In non-clumping regime

Run	Initial parameters		
	Z	St	α_M
Z0003St01	0.003	0.1	0
			0.01
Z0003St001	0.003	0.01	0
			0.01
Z002St01	0.02	0.1	0
			0.01
Z002St001	0.02	0.01	0
			0.01

Result of unstratified disk

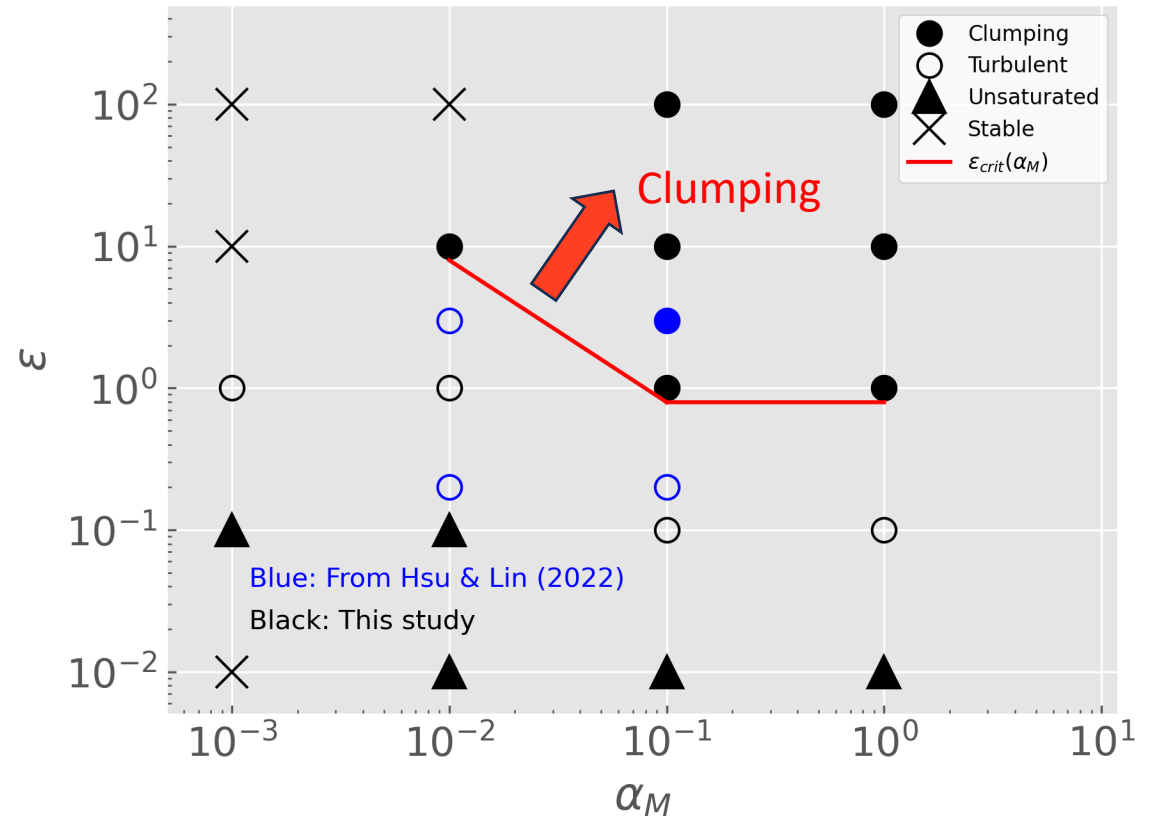


Result of unstratified disk

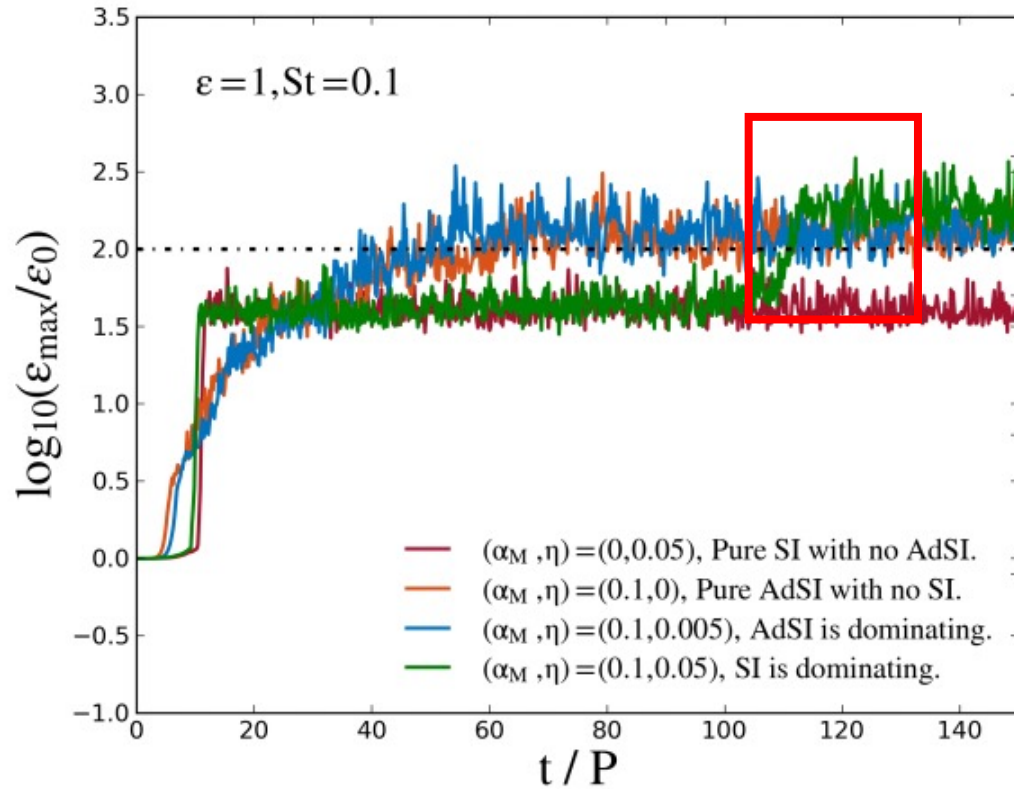


Clumping threshold

- $\epsilon > 1$ is needed for reaching clumping condition.
- If α_M is too small, no appreciable instability is produced.



Dust grow efficiency



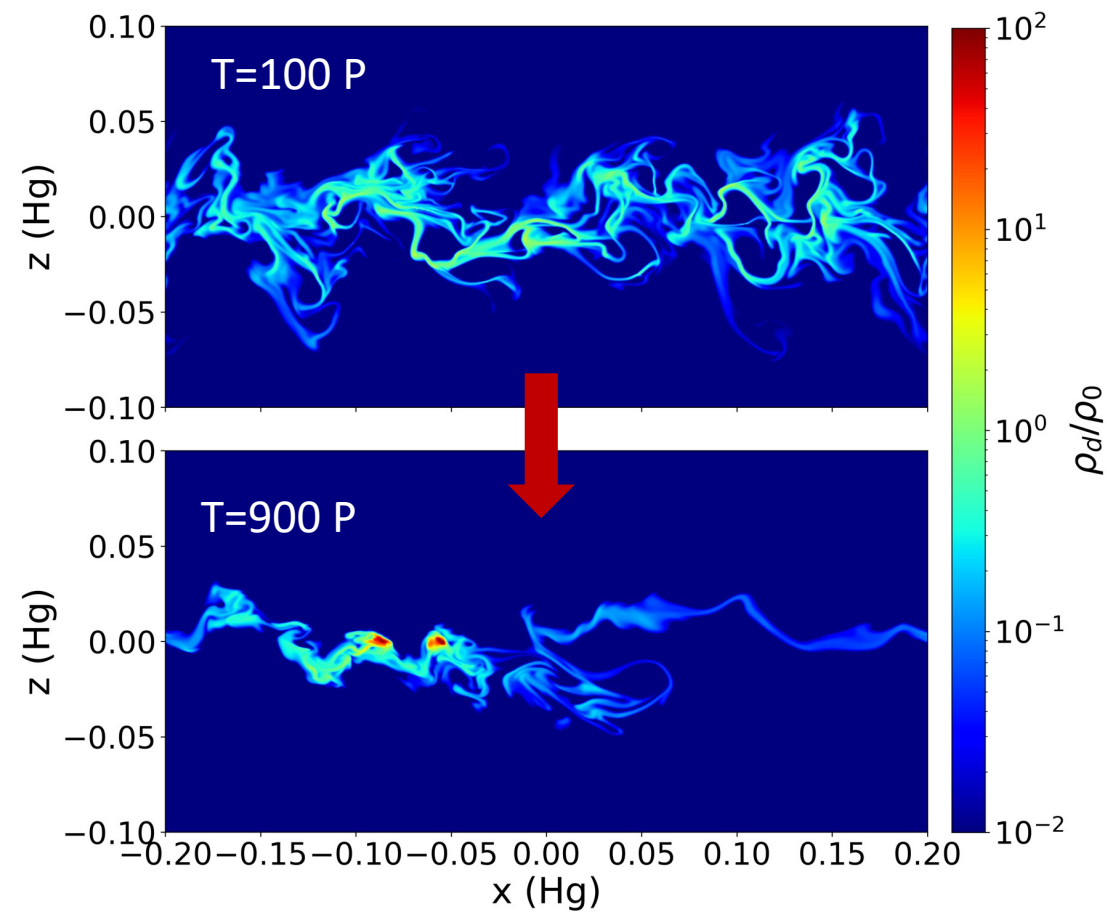
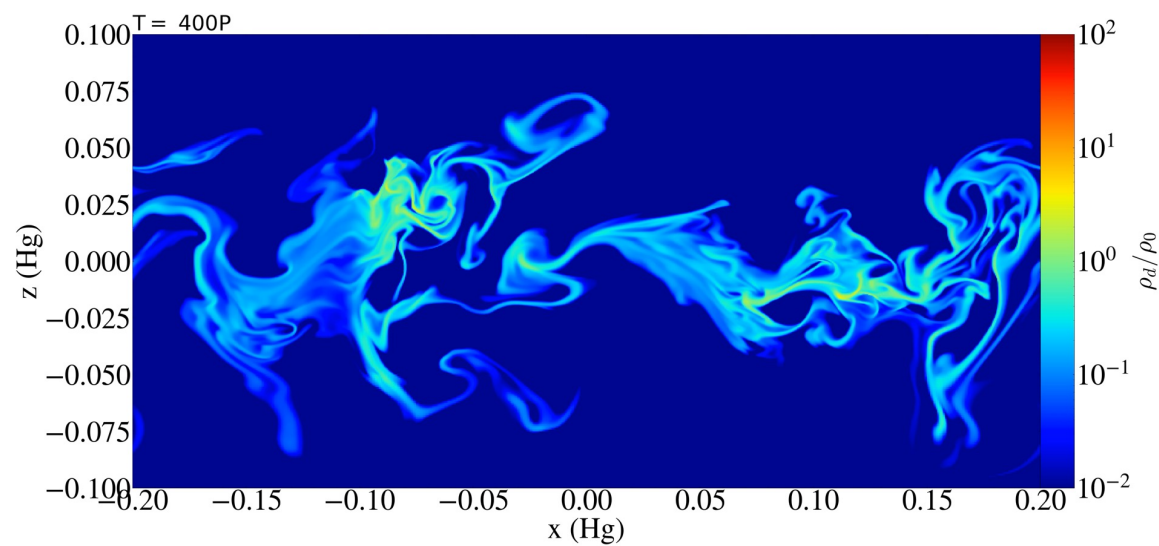
Azimuthal torque can provide relative drift between dust and gas, and result in radial drift.

Orange line, Blue line: AdSI itself can still accrete dust particles.

Red line: SI alone can accrete dust particles up to $\sim \rho_d/\rho_g = 32$.

Green line: AdSI can further help to concentrate dust to form dust clumps.

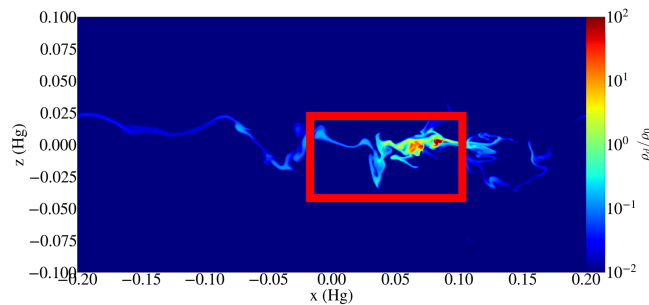
Result of stratified disk



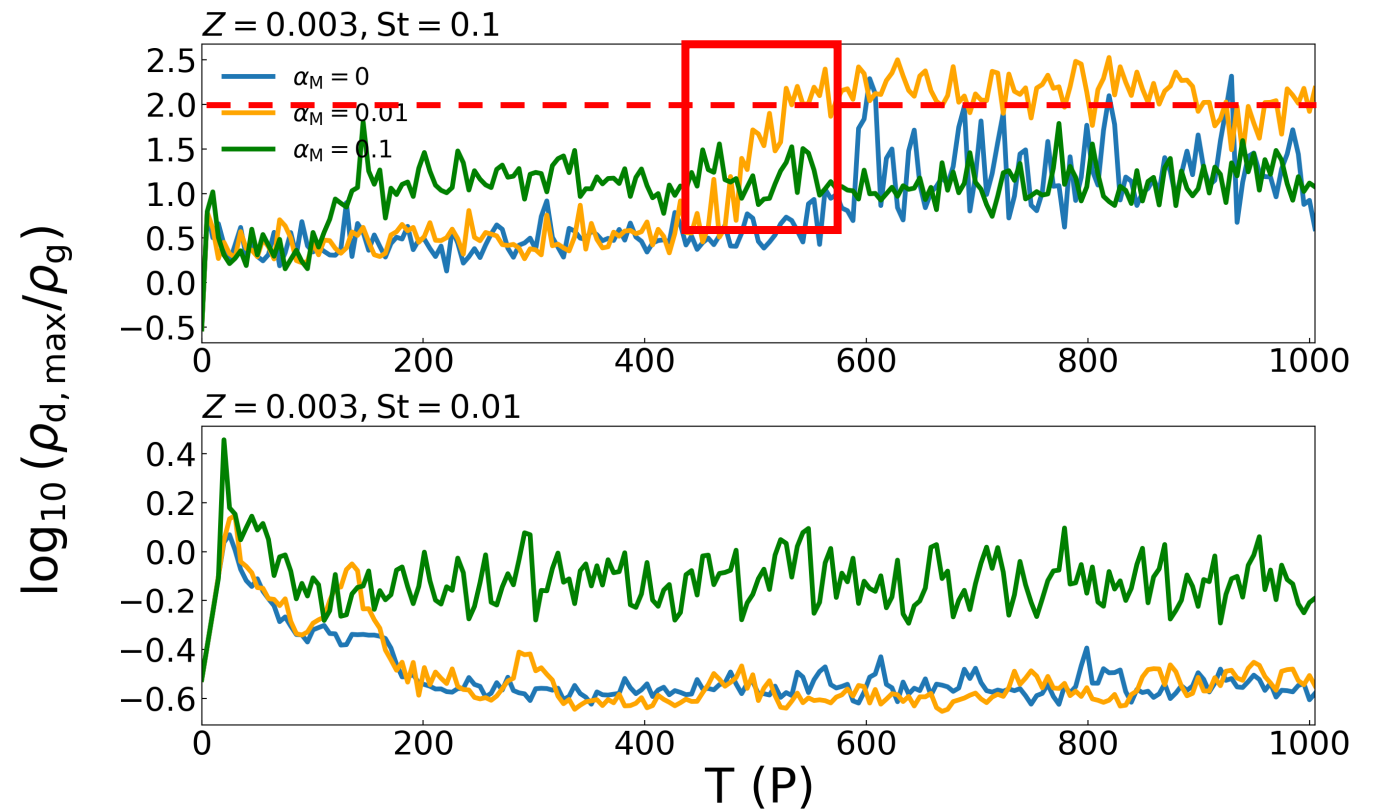
Run Z=0.003 (non-clumping regime)

Moderate azimuthal torque triggers dust concentration.

Excessively high torque stabilizes the disk, preventing further growth.



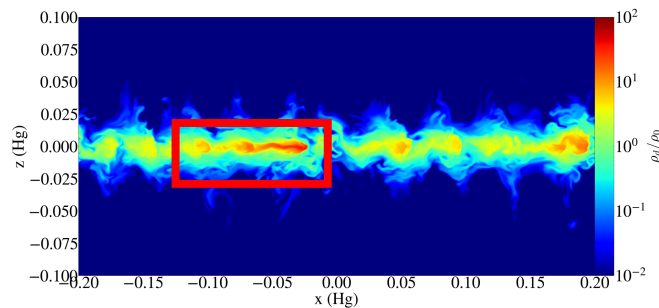
$Z=0.003, St=0.1, \alpha_M=0.01$



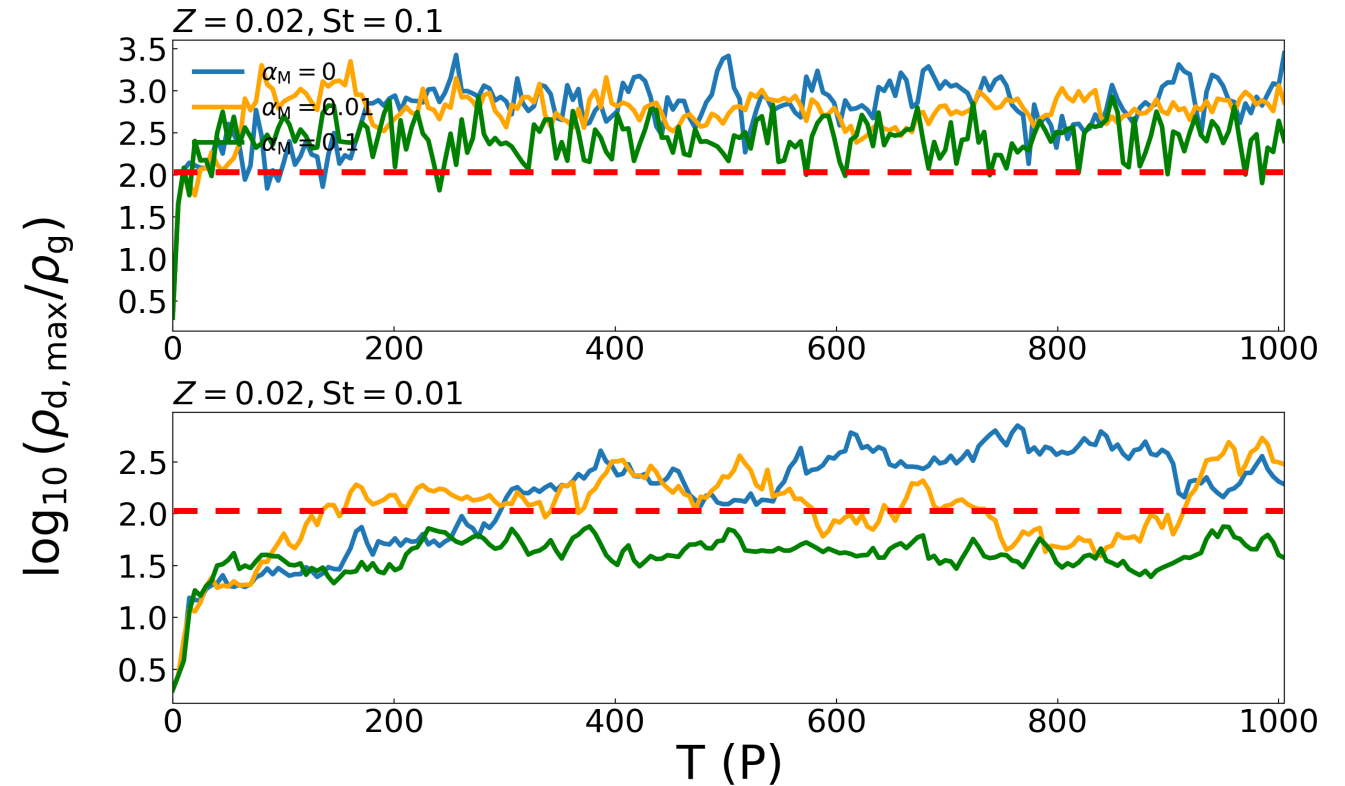
Run Z=0.02 (clumping regime)

α_M increase $\rightarrow$ suppress clumping efficiency.

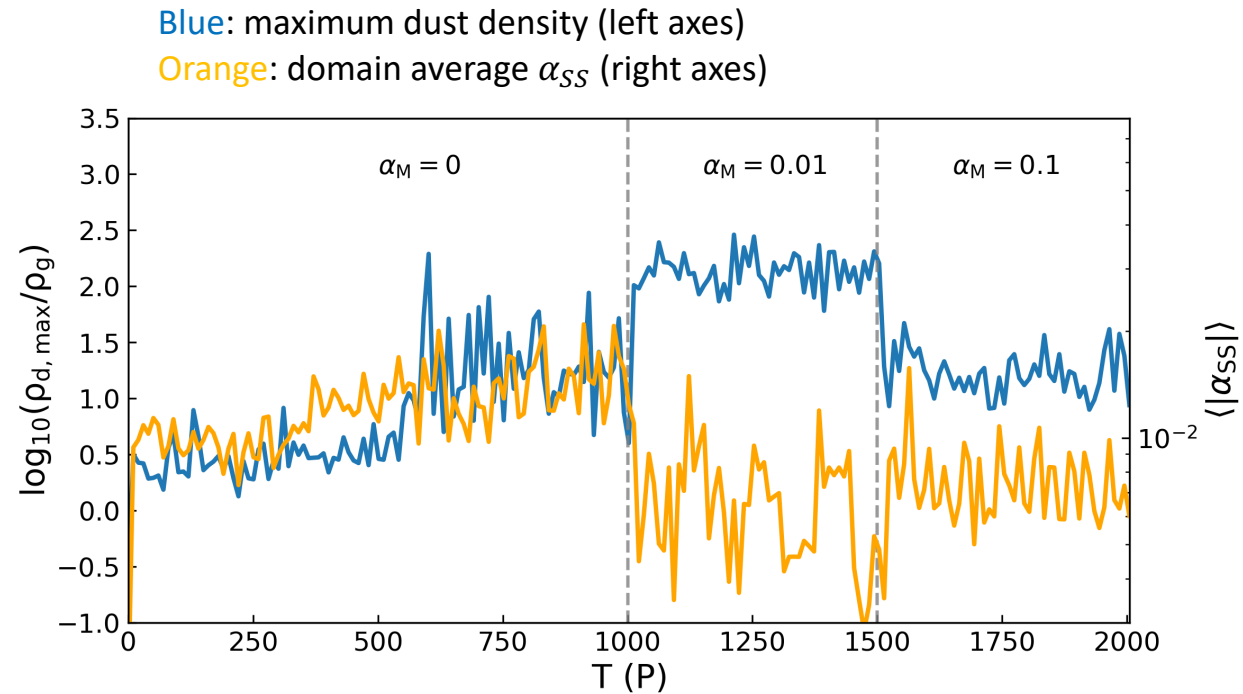
Excessive torque disrupts established clumps, causing densities to drop below the clumping threshold.



Z=0.02, St=0.01, $\alpha_M=0.1$



Turbulence measurement



$Z=0.003, St=0.1$

Restart at $T=1000$ P, $\alpha_M = 0 \rightarrow \alpha_M = 0.01$
at $T=1500$ P, $\alpha_M = 0.01 \rightarrow \alpha_M = 0.1$

Angular momentum flux: $\alpha_{SS} = \frac{\rho_g v_{gx} v_{gy}}{\rho_0 c_s^2}$

The localized reduction in ambient α_{SS} depends on α_M .

Summary

- Unstratified disk
 - critical dust-to gas ratio with roughly $\epsilon > 1$ allows dust to reach clumping condition.
 - The accretion flow generated by AdSI can further facilitate dust clumping in SI-driven accretion flows.
- Stratified disk
 - Azimuthal torque acts as a double-edged sword: it triggers clumping in dust-poor disks but suppresses it in dust-rich environments.
 - Too large α_M can stabilize disk, prevent the dust clump to form.